



## 2.1

# Change in Arithmetic and Geometric Sequences

## Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

### Student Information

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

### Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

### Readiness

1. Write one precise sentence explaining this principle: Arithmetic sequences have constant difference; geometric sequences have constant ratio.  
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2. Write one precise sentence explaining this principle: Sequence graphs are discrete because their domain is integer-valued.  
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3. Write one precise sentence explaining this principle: Indexing may start at 0 or 1; formulas must match the chosen index.  
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4. Write one precise sentence explaining this principle: A negative geometric ratio creates alternating signs.  
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**Core**

5. Write explicit and recursive formulas for 7, 12, 17, 22, ....
6. Write a formula for 160, 120, 90, ....
7. Which is larger for sufficiently large  $n$ :  $5 + 20n$  or  $5(1.2)^n$ ?
8. State what the formula  $a_n = a_0 + nd$  tells you and name any required restriction.
9. State what the formula  $a_n = a_1 + (n - 1)d$  tells you and name any required restriction.
10. State what the formula  $g_n = g_0r^n$  tells you and name any required restriction.

**Medium**

11. Classify and write an explicit formula for 4, 9, 14, 19, ....
12. Classify and write an explicit formula for 81, 27, 9, 3, ....
13. Classify and write an explicit formula for 5, -10, 20, -40, ....
14. Classify and write an explicit formula for -7, -3, 1, 5, ....

15. Classify and write an explicit formula for 2,6,18,54,....

**Hard**

16. Find the 40th term of the arithmetic sequence with  $a_7 = 19$  and  $a_{15} = 51$ .

17. Find  $g_1$  and  $r$  if  $g_4 = 54$  and  $g_7 = 1458$ .

18. An arithmetic sequence and a geometric sequence both begin 3,6. Find their third terms and explain divergence.

19. Write a recursive model for a balance starting 2000 QAR, increasing 4% then receiving 300 QAR at each period end.

**Difficult**

20. Find all arithmetic sequences satisfying  $a_3 + a_8 = 50$ .

21. Find a geometric sequence with positive terms such that  $g_2 = 12$  and  $g_5 = 324$ .

22. Prove that a nonconstant sequence cannot be both arithmetic and geometric for all  $n$ .

### Challenge / HOT

23. Construct an arithmetic sequence and geometric sequence that agree at exactly two specified indices.
24. Find all real  $r$  for which the geometric sequence  $1, r, r^2, \dots$  is also arithmetic for its first three terms.

### AP-Style Multiple Choice

25. Common difference of  $7, 3, -1, \dots$  is  
(A)  $-4$

- (B) 4  
(C) -3  
(D)  $1/4$
26. Common ratio of 5,-10,20,... is  
(A) -2  
(B) 2  
(C) -5  
(D)  $1/2$
27. Arithmetic explicit formula uses  
(A) powers of  $r$   
(B) a linear expression in  $n$   
(C) a logarithm  
(D) only recursion
28. Geometric graph is discrete because  
(A) outputs are positive  
(B) domain is integer indices  
(C) ratio is constant  
(D) it has an asymptote
29. If  $g_1 = 3, r = 2$ , then  $g_5 =$   
(A) 15  
(B) 24  
(C) 48  
(D) 96
30. Negative ratio causes  
(A) constant output  
(B) alternating signs  
(C) linear graph  
(D) undefined terms

### AP-Style Free Response

31. An auditorium has 18 seats in row 1 and 3 more seats in each next row. (a) Write recursive and explicit formulas. (b) Find row 25. (c) Find total seats in first 25 rows.

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32. A medicine amount is 500 mg initially and 72% remains each hour. (a) Write recursive and explicit

sequences. (b) Find amount after 8 hours. (c) Determine first hour below 50 mg.

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### Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

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Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.